INCH-POUND

MIL-R-83726/27M
5 January 2016
SUPERSEDING
MIL-R-83726/27L
13 June 2011

PERFORMANCE SPECIFICATION SHEET

# RELAYS, HYBRID, TIME DELAY, VOLTAGE SENSITIVE, DELAY ON DROP-OUT, CLASS A, TYPE V, HERMETICALLY SEALED (POTTED), SPDT, 2 AMPERE, TRACK MOUNTED

Inactive for new design after 23 June 1995.
No superseding document.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the relay described herein shall consist of this specification and MIL-PRF-83726.

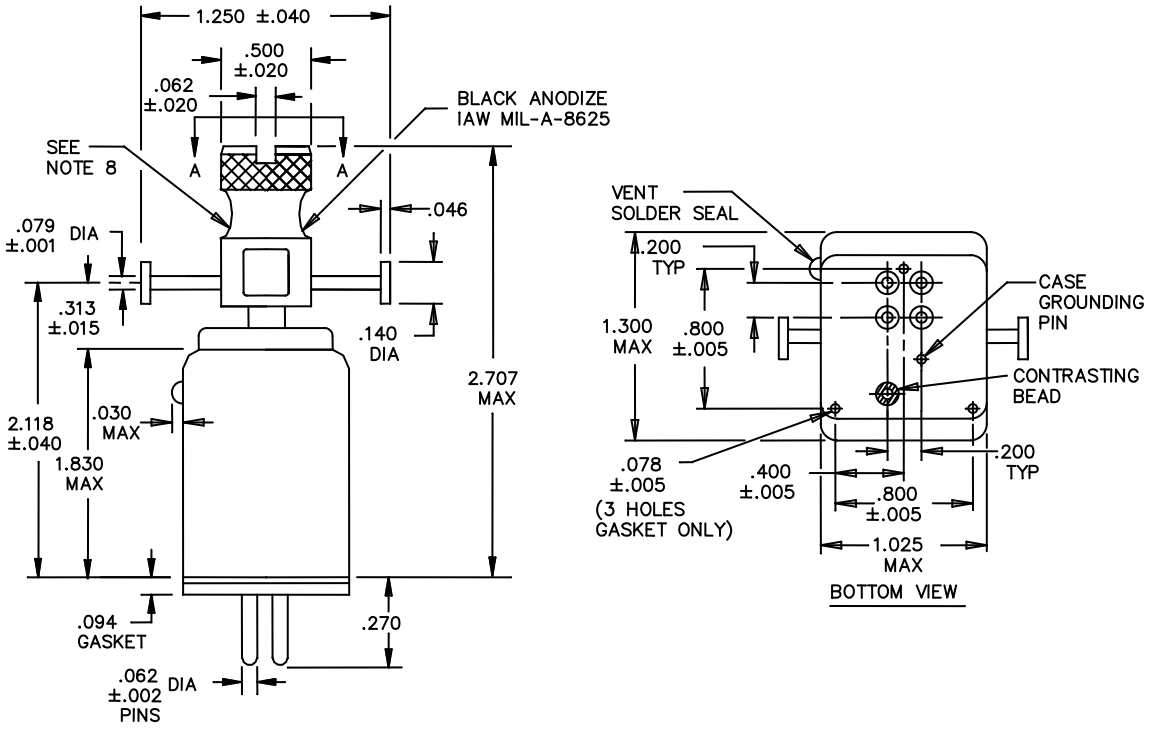
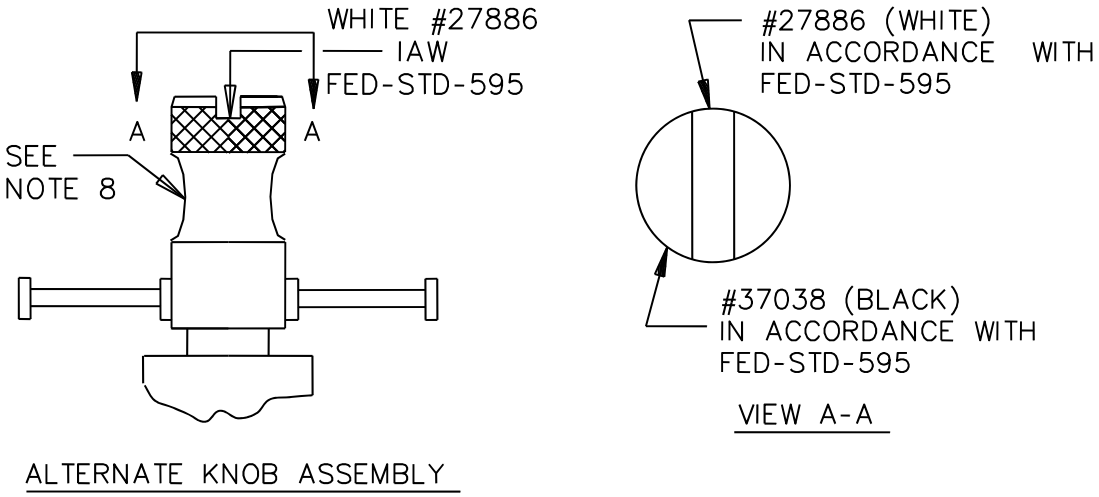



FIGURE 1. Dimensions and configuration.

AMSC N/A

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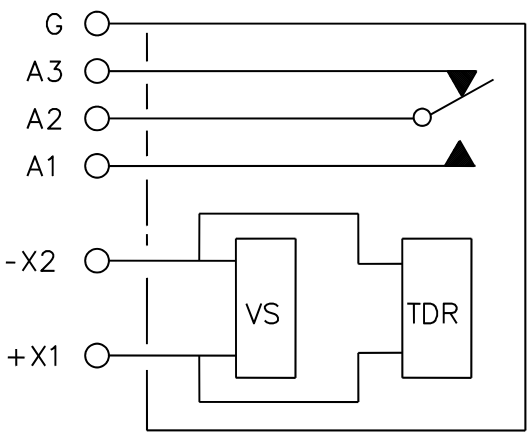




FIGURE 1. Dimensions and configuration - Continued.



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| Inches | mm | Inches | mm | Inches | mm |
|---|---|---|---|---|---|
| .001 | 0.03 | .062 | 1.57 | .400 | 10.16 |
| .002 | 0.05 | .078 | 1.98 | .500 | 12.70 |
| .005 | 0.13 | .079 | 2.01 | .800 | 20.32 |
| .015 | 0.38 | .094 | 2.39 | 1.025 | 26.04 |
| .020 | 0.51 | .140 | 3.56 | 1.250 | 31.75 |
| .030 | 0.76 | .200 | 5.08 | 1.300 | 33.02 |
| .040 | 1.02 | .270 | 6.86 | 1.830 | 46.48 |
| .045 | 1.14 | .313 | 7.95 | 2.118 | 53.80 |
| | | | | 2.707 | 68.76 |

NOTES:

1. Dimensions are in inches.
2. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
3. Metric equivalents are given for general information only.
4. Terminal numbers shall not appear on the relay header. There shall be affixed to the relay a legible circuit diagram which identifies each terminal location specified.
5. For mounting track, see MIL-DTL-12883/49. For bracket and socket assembly, see MIL-DTL-12883/51.
6. Gasket material: Methylvinyl silicone; shore hardness A40.
7. Terminal plating: Shall provide the operational, environmental, and interface characteristics to provide a reliable interconnect to gold plated contacts. Terminals except for the polarizing pin, shall be gold plated. One system for gold plating that may be used is ASTM B488, type 3, class 1.25 with a nickel underplate of 50 to 150 microninches thick. The gold plating system shall enable the product to meet the performance requirements of this specification and shall be approved by the qualifying activity.
8. Knob and post shall be nickel plated (100 microinches thick minimum) 303 stainless steel.

FIGURE 1. Dimensions and configuration - Continued.



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REQUIREMENTS:

Operating data:

Configuration: SPDT.

Life/load ratings: See table I.

Timing action: See table II.

TABLE I. Life/load ratings (relay case grounded).

| Type of load | Life (cycles) | Amperes 28 V dc |
|---|---|---|
| Resistive | 100,000 | 2.0 |
| Inductive | 50,000 | 0.5 |
| Lamp | 100,000 | 0.2 |
| Mechanical life (reduced current) | 400,000 | 0.5 |
| Intermediate current | 100,000 | 0.5 |

Contact drop:

Initial: 0.150 volt maximum.

After life tests: 0.175 volt maximum.

Minimum current: 0.050 ampere.

Contact bounce: See table II.

Overload:

DC: 8 amperes minimum.

AC: Not applicable.

Rupture:

DC: 10 amperes minimum.

AC: Not applicable.

Input data:

Duty ratings: Continuous.

Input voltage:

Maximum: See table II.

Rated: See table II.



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Voltage:

Operate: See table II.

Release: See table II.

Operate time: See table II.

Release time: See table II.

Maximum coil current: See table II.

TABLE II. Operating characteristics.

| Dash number | DC coil volts 1/ | | | Maximum dc coil | | Time-milliseconds-maximum | | |
|---|---|---|---|---|---|---|---|---|
| | Rated | Pickup 2/ | Dropout | Volts | Amps | Operate 3/ | Release delay 4/ | Contact bounce 3/ |
| -001 | 28 | 15.5 +0.5, -0.0 | 14.0 -0.0 5/ | 30 | 0.11 | 10 | 150 ±50 | 1.0 |
| -002 | | 25.75 ±0.25 | 24.75 ±0.25 | | | | 3500 ±500 | |
| -003 | | 21.0 +0.5, -0.0 | 20.5 +0.0, -0.5 | | | | | |
| -004 | | 15.5 +0.5, -0.0 | 14.0 -0.0 5/ | | | | 1000 ±200 | |

1/ Over total operating temperature range.
2/ Applies also to MIL-PRF-6106 high temperature and continuous current tests.
3/ With rated coil voltage.
4/ From dropout voltage to 8 volts. Below 8 volts, release delay time may be 0.
5/ Plus tolerance permits any value less than pickup voltage.

Electrical data:

Insulation resistance:

Initial: See table III.

After life: See table III.

Dielectric withstanding voltage (sea level):

Coil to case:

Initial: See table III.

After life: See table III.



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All other points:

Initial: See table III.

After life: See table III.

Dielectric withstanding voltage (80,000 feet):

Coil to case:

Initial: See table III.

After life: See table III.

All other points:

Initial: See table III.

After life: See table III.

Leakage current: See table III.

TABLE III. Electrical data.

| Minimum insulation resistance (megohms) | | Dielectric withstanding voltage (rms) | | | | |
|---|---|---|---|---|---|---|
| | | Sea level 1/ | | | 80,000 feet altitude 2/ | |
| Initial | After life | | Initial | After life | Initial | After life |
| 100 | 100 | Coil to case | 1050 | 1050 | 350 | 350 |
| | | All other points | 600 | 600 | 350 | 350 |
| | | Normal leakage current (rms) microamperes 3/ | | | | |
| | | Coil to case | 25 | 25 | | |
| | | All other points | 25 | 25 | | |

1/ Testing shall be performed with relay installed in socket and bracket assembly MIL-DTL-12883/51 and secured in mounting track MIL-DTL-12883/49.
2/ Relay shall not be installed in socket and bracket assembly MIL-DTL-12883/51.
3/ Leakage current at sea level.

Transients (applicable O -003 only): The time delay relay shall comply with MIL-STD-461 (applies to X1 and X2 terminals only; A1, A2, and A3 shall be wired but unloaded) and the requirements defined in method CE106, CS101, CS114, RE103, and RS103, except that the field intensity of RS103 shall be in accordance with table IV herein. Voltage spikes (transients of 500 microseconds or less duration) generated by the time delay relay and conducted on lines X1, X2 of the relay shall not exceed 42 volts. Voltages generated by the time delay relays shall be monitored on the control/power leads (+X1, X2) with an oscilloscope having the ability to detect, measure, and display voltage spikes of 0.25 to 1,000 volts amplitude and 10 nanoseconds to 50 microseconds duration. The oscilloscope shall be capable of retaining the display. The qualification test report shall include photographs of all oscilloscope-displayed spike voltages. Switches used to control the time delay relays shall not be EMI-suppressed in any way. During susceptibility testing, the relay shall not change state nor have the timing interval affected in any way by the EMI signals. The timing interval shall be considered to include the instants of turn-on and turn-off.



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TABLE IV. Field intensity levels.

| Frequency | RS103 (volts/meter) |
|---|---|
| 14 kHz to 200 MHz | 20 |
| 200 MHz to 450 MHz | 65 |
| 450 MHz to 1 GHz | 20 |
| 1 GHz to 12 GHz | 100 |

Environmental data:

Temperature range (operating): -70°C to +85°C. +171°C nonoperating, 10 minutes. +200°C, nonoperating, 1 minute.

Altitude: 80,000 feet.

Shock:

G level: 50 g's.

Duration: 10 milliseconds.

Contact opening: 10 microseconds, maximum.

Contact closing: 1 microsecond, maximum.

Vibration (sinusoidal):

G level: 30 g's.

Frequency range: 10 Hz to 3,000 Hz.

Vibration (random): Applicable in accordance with MIL-STD-202, method 214, test condition 1B.

Power specter density: 0.4 $g^2$/Hz.

Frequency range: 50 Hz to 2,000 Hz.

Duration: 15 minutes each plane.

Vibration scan:

Frequency range: 5 Hz to 2,000 Hz.

Intensity: 0.01 inch double amplitude or ±2 g's (whichever is less).

Acceleration: 15 g's.

Seal: Applicable to nonhermetically-sealed relays and hermetically-sealed (potted) relays.



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Physical data:

Dimensions and configuration: See figure 1.

Weight: 0.24 pound (.108 kilogram) maximum.

Finish: Orange 32246, FED-STD-595.

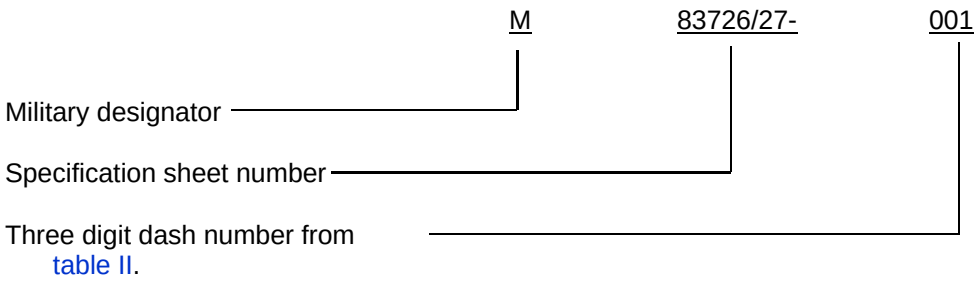
Marking: See MIL-PRF-83726. In addition, relays shall be marked with the Electrostatic Discharge Sensitive (ESDS) identifier as specified in MIL-STD-1285.

ESDS protection program: The manufacturer shall establish and maintain an ESD control program in accordance with MIL-STD-1686 for mission critical equipment. Evidence of such compliance shall be verified by the qualifying activity of this specification as a prerequisite for qualification and continued qualification. This program shall be documented by an ESD control plan that must be under document control. As a minimum, this plan must address the identification of ESDS sub-components and end items, facilities, training, design protection, handling procedures, marking, cleaning, preservation, packaging, and quality assurance. A model ESD control program is available from the qualifying activity and may be used as a guideline. Further guidance for ESD control is available from the Electrostatic Discharge (EOS/ESD) Association and the Electronics Industry Association (EIA). This requirement is applicable to all manufacturers who handle ESDS component parts and materials in the relay manufacturing or testing process. This requirement is not limited to manufacturers qualifying ESDS end items.

ESDS verification: As a part of qualification or qualification after redesign, ESD testing shall be done in accordance with method 3015 of MIL-STD-883 modified to test at 16,000 volts. Testing at lower voltage levels is not required. This testing shall be accomplished as part of the group III for qualification inspection and as part of the group C inspection.

ESDS preservation and packaging: Relays shall be preserved and packaged in such a manner as to ensure that the integrity of ESD sensitive relays is not diminished. ESD sensitive relays shall be preserved and packaged in accordance with the requirements of MIL-STD-1686.

Part or Identifying Number (PIN): Consists of the prefix M83726/27-, a three digit dash number (expressed in milliseconds):




Example: M83726/27-001



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Quality assurance provisions: Group B and group C testing are not required. The manufacturer shall notify the qualifying activity in the event of any design or construction changes, and shall impose additional testing requirements as necessary.

The Qualified Products List (QPL) associated with this inactive for new design specification will be maintained until acquisition of the product is no longer required, whereupon the specification and the QPL will be canceled.

NOTE: As of 15 June 1999, MIL-PRF-83726 no longer specifies Quality Levels, but existing order configurations may still include them. Relays with a "W" quality level indicator at the end are interchangeable with relays without a quality level letter.

Referenced documents. In addition to MIL-PRF-83726, this document references the following:

| | | | |
|---|---|---|---|
| FED-STD-595/27886 | FED-STD-595/37038 | FED-STD-595/32246 | MIL-DTL-12883/51 |
| MIL-DTL-12883/49 | MIL-PRF-6106 | MIL-A-8625 | MIL-STD-202-214 |
| MIL-STD-461 | MIL-STD-883 | MIL-STD-1285 | MIL-STD-1686 |
| ASTM B488 | | | |

Changes from previous issue: The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

Custodians:
Navy - EC
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

Review activity:
Air Force – 99

(Project 5945-2016-007)

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